

NENYASHA MADYAVANHU

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Portfolio: <https://nenyasha.pythonanywhere.com/> | GitHub: <https://github.com/Dei-gratia>

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SUMMARY

A highly motivated and results-driven Computer Science and Engineering professional with a solid foundation in data science, cloud computing, and software development. Skilled in machine learning, full-stack development, and applied research, with peer-reviewed publications and a strong passion for technological innovation and building and maintaining cutting-edge solutions.

CORE COMPETENCIES AND SKILLS

- **Development & Programming:** Python, Java, JavaScript, C/C++, SQL/NoSQL, Django, React.js, RESTful APIs
- **Data Science & AI:** Machine Learning, Artificial Intelligence, Predictive Modelling, Forecasting, Optimization, Data Analysis, Scikit-learn, TensorFlow, Pandas, NumPy, Matplotlib, Seaborn, Computer Vision, NLP
- **Tools & Platforms:** Git, GitHub, Power BI, Excel, Linux, MATLAB, CloudSim, Azure ML studio, Tableau, Weka
- **Technical Courses:** Operating Systems, Computer Networks, Database Management Systems, Data Structures & Algorithms, Information Security, Linux and Shell Scripting, Data Warehouse and Mining, Big Data, Cloud computing, DevOps, Business Analytics
- **Research:** Published research on AI-driven virtual machine consolidation, cloud resource optimization, energy efficiency, and predictive modelling
- **Languages:** English (Fluent), Shona (Native)

EDUCATION

Master of Engineering in Computer Science and Engineering (CGPA-7.70/10)

Chandigarh University, India — 2024

- Conducted research on AI-driven optimization of energy and resource usage in cloud data centers
- Published 2+ peer-reviewed research papers

Bachelor of Technology in Computer Science and Engineering (CGPA-8.87/10)

APG Shimla University, India — 2022

PUBLICATIONS

1. N. Madyavanhu and V. Kumar, "Multi-resource balance Virtual Machine Consolidation Based on Extreme Learning Machine and Improved Grey Wolf Optimizer," in *2024 Sixth International Conference on Computational Intelligence and Communication Technologies (CCICT)*, Sonapat, India, 2024 pp. 183-192. doi: 10.1109/CCICT62777.2024.00039
2. Madyavanhu, N., Kumar, V. Utilizing multi-population ant colony system and exponential grey prediction model for multi-objective virtual machine consolidation in Cloud Data Centers. *Int. j. inf. tecnol.* (2024). Doi:10.1007/s41870-024-01949-0
3. N. Madyavanhu, "Enhancing Resource Utilization and Energy Efficiency in Cloud Data Centres: An Overview of VM Consolidation Techniques." *International Conference on Advances in Smart Sensor, Signal Processing, and Communication Technology*, March 2024. In Press.

WORK EXPERIENCE

Full Stack Development Intern (OceanaTech | Jan 2022 - July 2022)

- Built and deployed web applications using Django, Flask, React, and SQL/NoSQL databases. Developed REST APIs and Python automation scripts for data workflows. Collaborated in Agile teams and performed debugging, testing, and cloud deployments.

CERTIFICATIONS & TECHNICAL TRAINING

Microsoft Certified: Azure Data Scientist Associate: Microsoft | Jun 2023

Google Africa Developer Training program: Google Developers | May 2022

- Completed 6 months of certified training in Android mobile application development, covering Kotlin, advanced Android concepts, real-time databases, and app deployment, with multiple projects and a final project featured as one of the best.

Android App Development Trainee: Internshala Trainings | Dec 2021

- Completed 8 weeks of certified training covering Android basics, Kotlin, advanced functionalities, and a capstone project.

Web Development Trainee: Internshala Trainings | Oct 2021

- Completed 8 weeks of certified training covering HTML, CSS, Bootstrap, DBMS, PHP, JavaScript, React, and a final project.

REFERENCES

Dr. Vaneet Kumar

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Dr. Paurav Goel

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